CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

7.1 Progressive Waves

Syllabus Section: 7. Waves **Paper Type:** Structured Questions (Paper 2) **Total:** 6 Questions (59 Marks)

Question 1 Easy [6 marks]

Part (a) [1 mark]

State the type of wave for which the oscillations are perpendicular to the direction of energy transfer.

Mark Scheme (1 marks):

The oscillations are perpendicular to the direction of energy transfer for...

- Transverse waves; *[1 mark]*

[Total: 1 mark]

If you need a handy way to remember which is which



For transverse waves, the oscillations are at right angles (perpendicular) to the direction of energy transfer and wave motion.

Longitudinal waves have compressions and rarefactions and so need particles to propagate.

Part (b) [2 marks]

State two objects that can be used in the classroom to demonstrate wave motion.

Mark Scheme (2 marks):

Two objects that can be used in the classroom to demonstrate wave motion are...

*Any **two** from:*

- Rope; *[1 mark]*
- Spring; *[1 mark]*
- Ripple tank; *[1 mark]*

[Total: 2 marks]

You are expected to be familiar with all three.

Part (c) [1 mark]

Fig. 1.1 shows a waveform consisting of compressions and rarefactions.

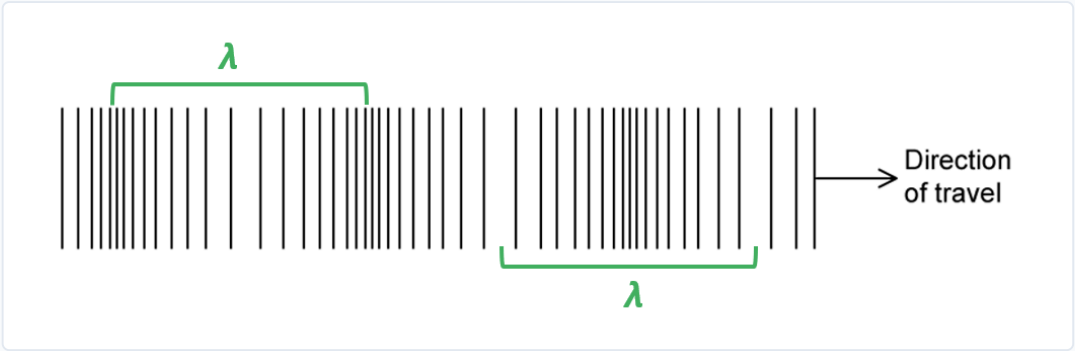


Fig. 1.1

Annotate the diagram to show one complete wavelength, and label it λ.

Mark Scheme (1 marks):

A wavelength on a longitudinal wave is as follows:



- A line spanning from the centre of one compression (**or** rarefaction) to the same point on the next compression (**or** rarefaction); *[1 mark]*

[Total: 1 mark]

Part (d) [2 marks]

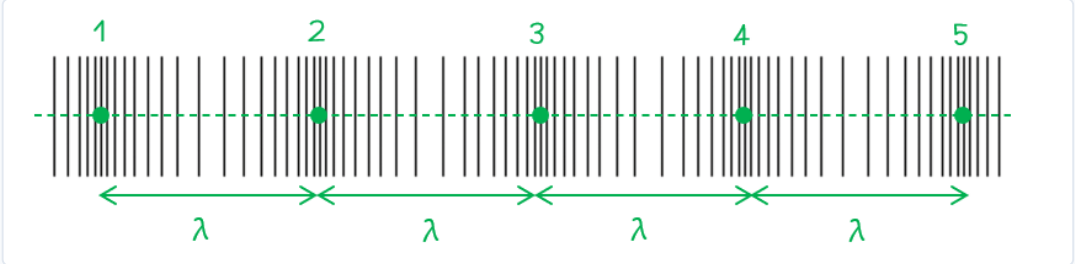
The compression wave in Fig. 1.1 is a progressive wave in which 5 compressions pass a fixed point every 2 seconds.

Determine the frequency of the wave.

Mark Scheme (2 marks):

Determine the number of complete wavelengths:

- Over 5 compressions, there are 4 complete wavelengths



Determine the frequency of the wave:

- $\text{Frequency} = \frac{\text{number of complete wavelengths}}{\text{time taken}}$
- $f = \frac{4}{2}$ *[1 mark]*
- $f = 2 \text{ Hz}$ *[1 mark]*

[Total: 2 marks]

Be careful with the wording of the question. 5 compression points pass a fixed point. The first compression point passing triggers the timing to start. When the second compression point passes, this means that one full wavelength has passed.

Question 2 Easy [12 marks]

Part (a) [2 marks]

Fig. 1.1 shows a displacement-distance graph for a travelling wave.

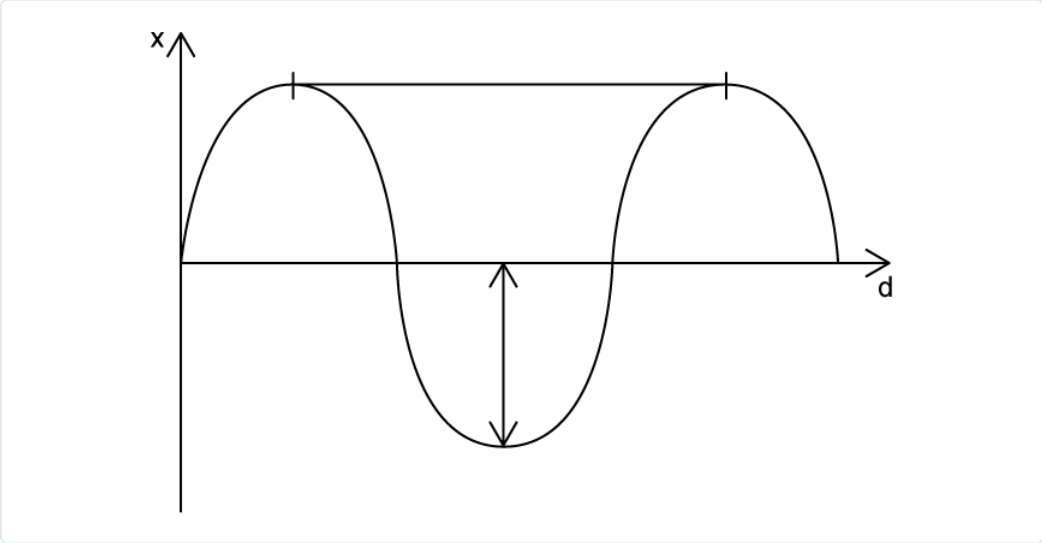


Fig. 1.1

On Fig. 1.1, add labels to show:

(i) the wavelength

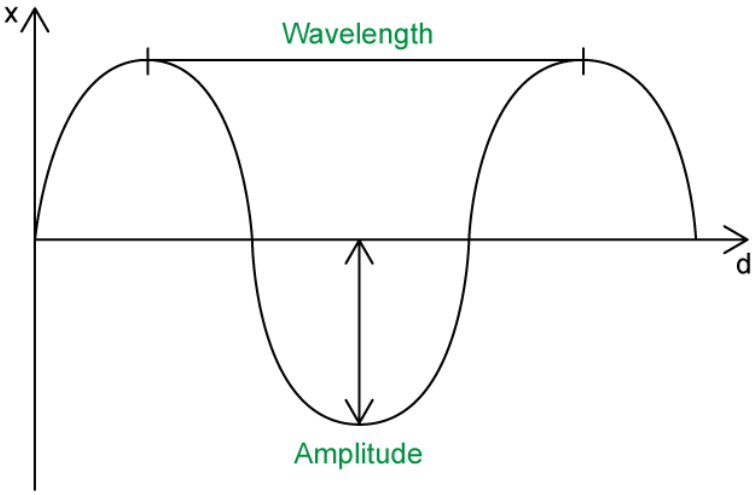
[1]

(ii) the amplitude

[1]

Mark Scheme (2 marks):

A correctly labelled diagram scoring two marks:



(i) Label the wavelength:

- Wavelength / λ correctly labelled from crest to crest; *[1 mark]*

(ii) Label the amplitude:

- Amplitude / x_0 correctly labelled from point of zero disturbance to maximum displacement; *[1 mark]*

[Total: 2 marks]

Writing the words, or giving the correct symbols would get you full marks.

Part (b) **[5 marks]**

Match the key word to its correct definition

Period	Time taken for a point on a wave to undergo one complete oscillation
Amplitude	Number of complete oscillations per second
Frequency	Distance travelled by a wave per unit time
Wavelength	Maximum displacement of a wave from its equilibrium position
Wave speed	Distance between the same point on two consecutive waves

Mark Scheme (5 marks):

An answer scoring 5 marks:

Period	Time taken for a point on a wave to undergo one complete oscillation	[1 mark]
Amplitude	Number of complete oscillations per second	[1 mark]
Frequency	Distance travelled by a wave per unit time	[1 mark]
Wavelength	Maximum displacement of a wave from its equilibrium position	[1 mark]
Wave speed	Distance between the same point on two consecutive waves	[1 mark]

[Total: 5 marks]

Key words and definitions are extremely important in every topic. In extended answer questions, you will need to use key words correctly to score the marks.

Part (c) [1 mark]

Fig. 1.2 shows a displacement-time graph for an oscillating object.

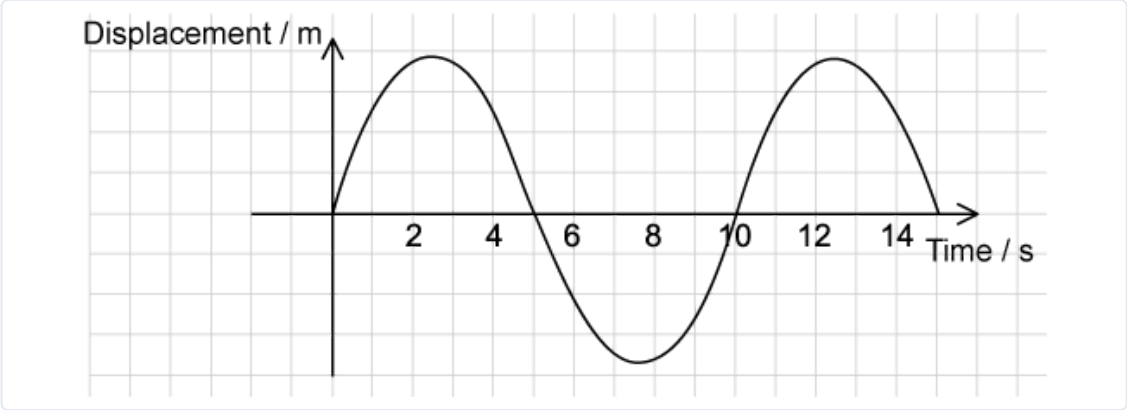


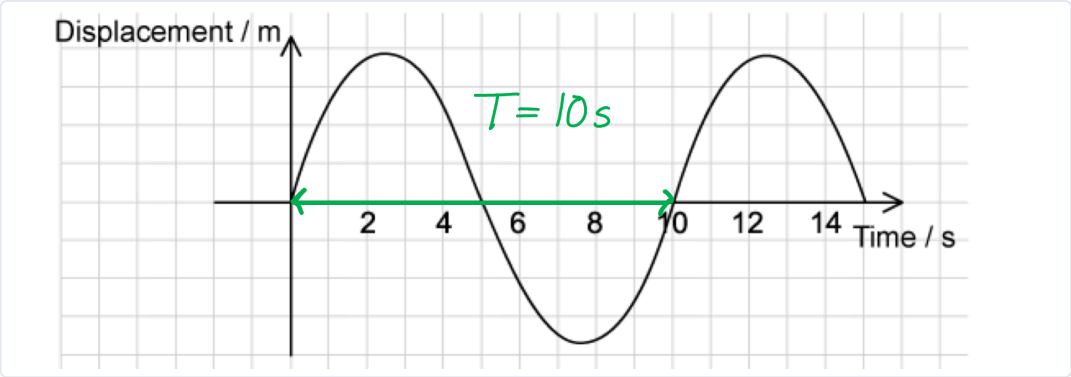
Fig. 1.2

Determine the time period *T* for this oscillation

Mark Scheme (1 marks):

Determine the time period *T* from the graph:

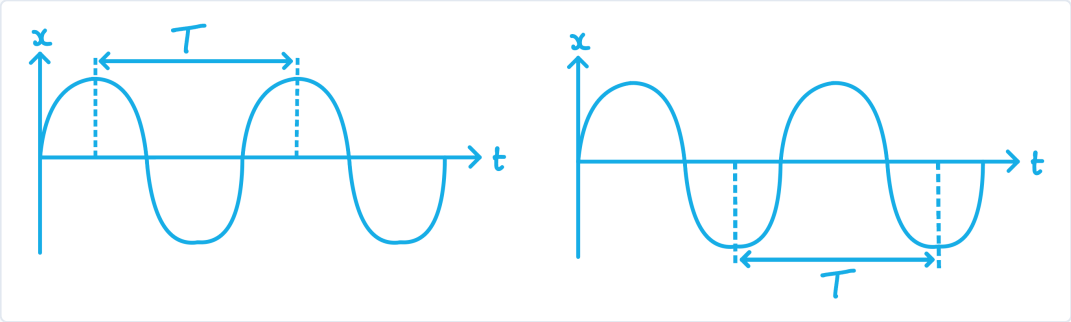
- Measure the time between the same point on two consecutive waves



- Correctly determine the period $\underline{T = 10\ s}$; *[1 mark]*

[Total: 1 mark]

Remember that the time period on a displacement-time graph is the distance between one point on one wave and ***the same point*** *on the next wave. You can measure the period from any point, such as crest to crest, or trough to trough.*



Always chose the point that will give you the most accurate reading. In this example, the most accurate point to measure from was time t = 0.

Part (d) **[4 marks]**

The oscillation shown in Fig. 1.2 has a wavelength λ of 5 m.

Calculate:

(i) the frequency f of the oscillation

[2]

(ii) the wave speed v.

[2]

Mark Scheme (4 marks):

List the known quantities:

- Time period, T = 10 s (from part c)

- Wavelength, λ = 5 m

(i) Calculate the frequency *f*:

- $f = \frac{1}{T}$ *[1 mark]*
- $f = \frac{1}{10}$ = 0.1 Hz *[1 mark]*

(ii) Calculate the wave speed *v*:

- $v = f\lambda$ *[1 mark]*
- v = 0.1 × 5 = 0.5 m s^{-1} *[1 mark]*

[Total: 4 marks]

Question 3 Medium [5 marks]

Part (a) [5 marks]

Circular water waves are produced by vibrating dippers at points P and Q, as illustrated in Fig. 4.1.

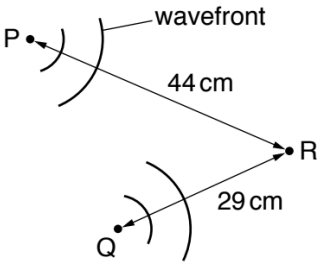


Fig. 4.1 (not to scale)

The waves from P alone have the same amplitude at point R as the waves from Q alone. Distance PR is 44 cm and distance QR is 29 cm.

The dippers vibrate in phase with a period of 1.5 s to produce waves of speed 4.0 cm s^{-1}.

(i) Calculate the wavelength of the waves.

wavelength = ... cm [2]

(ii) Calculate the path difference at point R of the waves from P and Q.

Give your answer in terms of the wavelength λ of the waves.

path difference = ... λ [1]

(iii) Describe the motion, if any, of the water particles at point R.

Explain your answer.

[2]

Mark Scheme (5 marks):

List the known quantities:

- Wave speed, v = 4.0 cm s^{-1}
- Time period, $T = \frac{1}{f}$ = 1.5 s
- Distance PR = 44 cm
- Distance QR = 29 cm

i) Use the wave equation to determine wavelength, λ:

- $v = f\lambda$

- $\lambda = \frac{v}{f}$ *[1 mark]*
- λ = 4.0 × 1.5 = 6.0 cm *[1 mark]*

ii) Calculate the path difference at point R of the waves from P and Q:

- The difference between distances PR and QR = 44 − 29 = 15 cm
- Number of wavelengths in 15 cm = $\frac{15}{6} = \frac{5}{2}$ = 2.5
- Path difference = 2.5λ or $\frac{5}{2}\lambda$ *[1 mark]*

iii) Describe the motion, if any, of the water particles at point R:

- The path difference is an odd number of half wavelengths **OR** the waves have a phase difference of 180°; *[1 mark]*
- (This results in destructive interference, hence) the particles are at rest / have no motion; *[1 mark]*

[Total: 5 marks]

- *Spotting that* $\lambda = \frac{v}{f} = v \times \frac{1}{f} = v \times T$ *can save you some time in part (i)*
- *Remember to state* ***path difference*** *as a* ***multiple of wavelength*** *for part (ii)*

Question 4 Medium [9 marks]

Part (a) [2 marks]

Fig. 1.1 represents a progressive wave travelling from left to right on a stretched string.

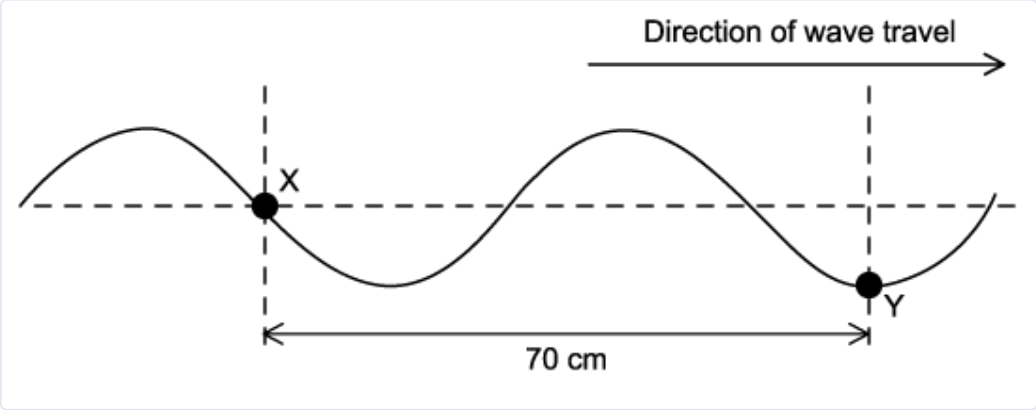


Fig. 1.1

The frequency of the wave is 30 Hz. Calculate the speed of the wave.

Mark Scheme (2 marks):

To calculate the speed:

Determine the wavelength:

- **XY** = 70 cm = $1\frac{1}{4}\lambda$ (or $\frac{5}{4}\lambda$)
- $\lambda = 70 \times \frac{4}{5}$ = 56 cm = 0.56 m *[1 mark]*

Wave speed calculation:

- $v = f\lambda$
- v = 30 × 0.56
- v = 16.8 m s^{-1} *[1 mark]*

[Total: 2 marks]

Part (b) [2 marks]

State the phase difference between points **X** and **Y** on the string, giving an appropriate unit.

Mark Scheme (2 marks):

The phase difference is:

- 90 / 450 *[1 mark]*
- ° / degrees *[1 mark]*

OR

- $\frac{\pi}{2}$ or $\frac{3\pi}{2}$ or $\frac{5\pi}{2}$ *[1 mark]*
- Radians / rad / **r** *[1 mark]*

Do not accept R, Rad etc.

[Total: 2 marks]

It is much better to represent the phase in terms of an angle instead of the number of wavelengths, unless you are asked to do so.

Part (c) **[3 marks]**

Describe how the vertical displacement of point **X** on the string varies in the next period.

Mark Scheme (3 marks):

To describe how the vertical displacement of point **X** on the string varies in the next period:

First $\frac{1}{4}$ cycle:

- Displacement of X will be **positive** (or '**up**') to a maximum positive displacement; *[1 mark]*

Next $\frac{1}{2}$ cycle:

- **Downwards** to equilibrium position ($\frac{1}{4}$ cycle) **and** maximum negative displacement ($\frac{1}{4}$ cycle); *[1 mark]*

Last $\frac{1}{4}$ cycle:

- **Upwards** to equilibrium position / zero displacement; *[1 mark]*

If the motion of X is 180 degrees out of phase with the one shown then a maximum of two marks can be gained (i.e. X initially moving downwards)

Accept 0.008 s for $\frac{1}{4}$ cycle and 0.017 for $\frac{1}{2}$ cycle

[Total: 3 marks]

*Visualising the subsequent motion of a point on the wave is tricky. The wave is travelling to the right hand side, so it's subsequent motion will be the wave motion drawn **behind** it.*

Part (d) **[2 marks]**

Determine the phase difference between the current position of **X** and the position of **X** 0.0825 s later.

Mark Scheme (2 marks):

List the known quantities:

- Frequency, f = 30 Hz (From part (a))

To determine the phase difference between the current position of **X** and the position of **X** 0.0825 s later:

Calculate the time period of 1 wave cycle:

- 1 period = $\frac{1}{f}$ = $\frac{1}{30}$ = 0.033 s

Determine the number of wave cycles in 0.0825 s:

- 0.0825 s is $\frac{0.0825}{0.033}$ = 2.5 complete cycles *[1 mark]*

Determine the phase difference over 2.5 cycles:

- The phase difference of 1 cycle = 360 °
- So, the phase difference of ½ cycle = 180°
- (Therefore, there is a) phase difference of 180° / π radians / anti-phase (between the two positions of **X**) *[1 mark]*

[Total: 2 marks]

Remember that a phase difference of 360 degrees is a complete 1 wavelength cycle, so there is actually no phase difference.

Question 5 Hard [14 marks]

Part (a) [3 marks]

Fig. 1.1 shows a vertical cross-section through a water wave moving from left to right.

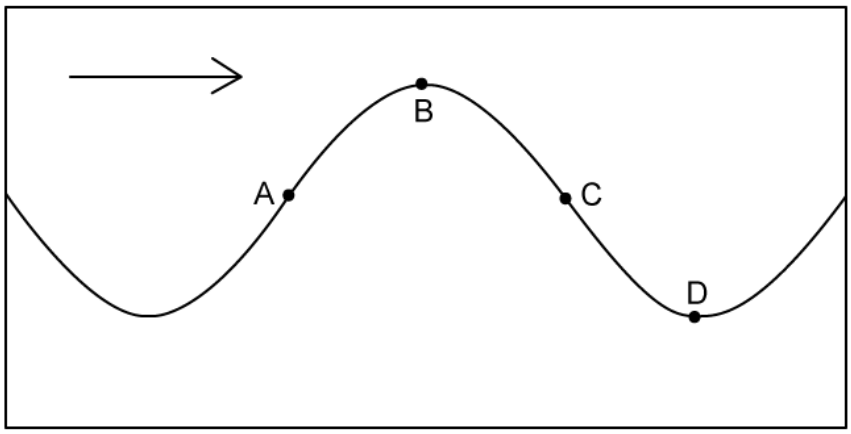


Fig. 1.1

Deduce the point at which the water is moving upwards at maximum speed and explain your reasoning.

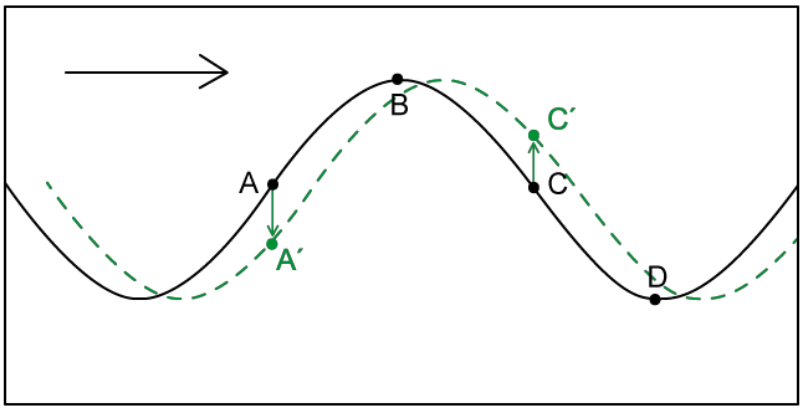
Mark Scheme (3 marks):

State the position on the wave where it moves upwards with maximum speed:

- C; *[1 mark]*

Explain the motion of the progressive wave:

- Since this is a progressive wave moving left to right, position C will move upward; *[1 mark]*



- Point C is at the equilibrium / undisturbed position / mid-point on the wave

  AND

  (Therefore) it is moving at maximum speed; *[1 mark]*

[Total: 3 marks]

Part (b) **[4 marks]**

While a microphone is connected to a cathode-ray oscilloscope, a student strikes a tuning fork nearby. The sound wave produced by the tuning fork is displayed as a trace on the screen.

Fig. 1.2 shows the resulting trace. The time-base and gain are given on the screen.

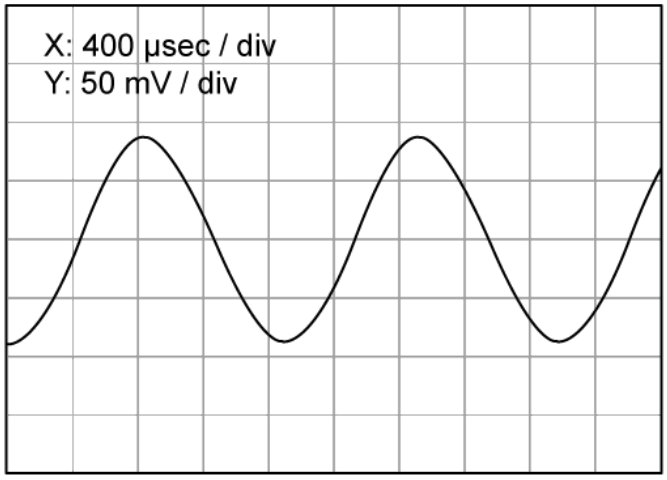


Fig. 1.2

Use Fig. 1.2 to calculate the wavelength of the sound wave produced.

(The speed of sound is approximately 330 m s^{-1}).

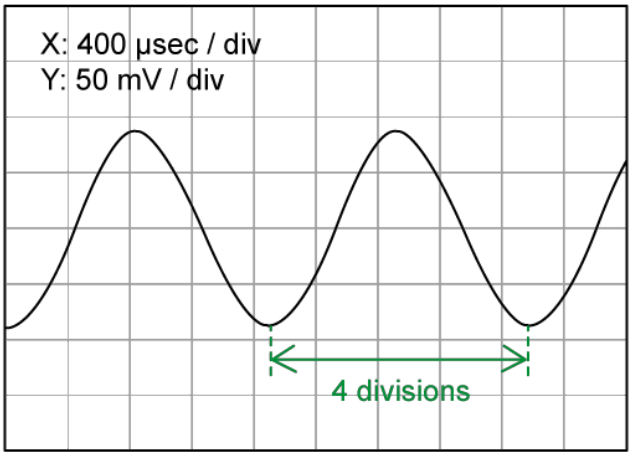
Mark Scheme (4 marks):

List the known quantities:

- Wave speed, v = 330 m s^{-1}

Determine the time period of the oscilloscope trace using the time-base:

- The time-base on the X-axis is set to a scale of 400 µs / div
- Therefore, each division represents 400 × 10^{-6} s
- One time period is 4 divisions (shown below); *[1 mark]*



- Therefore, time period (one full cycle), $T = 4 \times (400 \times 10^{-6}) = 1.6 \times 10^{-3}$ s *[1 mark]*

Calculate the frequency of the wave:

- $f = \frac{1}{T}$
- $f = \frac{1}{1.6 \times 10^{-3}} = 625$ Hz *[1 mark]*

Use the wave speed equation to calculate the wavelength of the sound wave:

- $v = f\lambda$
- $330 = 625 \times \lambda$
- $\lambda = \frac{330}{625} = 0.53$ m *[1 mark]*

[Total: 4 marks]

Part (c) **[3 marks]**

The student's lab partner strikes another identical tuning fork near the microphone, so that both sound wave signals can be overlaid on the oscilloscope trace. Fig. 1.3 shows the new trace.

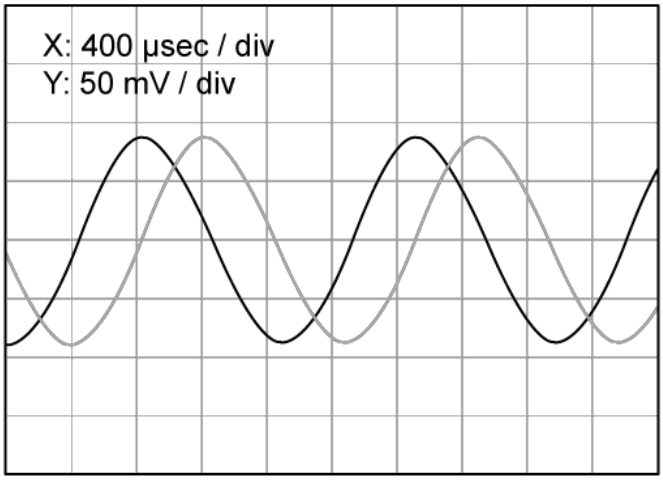


Fig. 1.3

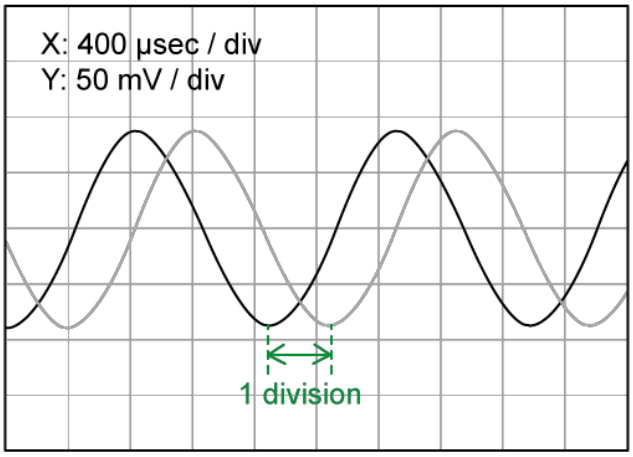
Using Fig. 1.3, determine the phase difference between the two signals as shown on the oscilloscope trace.

Give a suitable unit with your answer.

Mark Scheme (3 marks):

Determine the displacement between the two signals in terms of number of divisions:

- The two signals are displaced by 1 division; *[1 mark]*



Use the displacement to calculate a phase difference:

- If the two signals were in phase, the displacement would either be 0 divisions or 4 divisions
- Therefore, 1 division of displacement represents 25% of phase difference
- Phase difference = $\frac{1}{4} \times 2\pi = \frac{1}{2}\pi$

  OR

  Phase difference = $\frac{1}{4} \times 360 = 90$ *[1 mark]*

State the unit for phase difference:

- Phase difference = $\frac{1}{2}\pi$ rad

  OR

  Phase difference = 90º *[1 mark]*

[Total: 3 marks]

Part (d) **[4 marks]**

The oscilloscope trace is known as a 'time-view' of the sound wave because it shows the displacement of an oscillating signal against time. One of the students makes a time-view sketch as shown in Fig. 1.4, including a label *M*.

An oscillating signal can also be shown in terms of a 'space-view', which would show the displacement of the signal against its position. The other student makes a space-view sketch as shown in Fig. 1.5, including a label *N*.

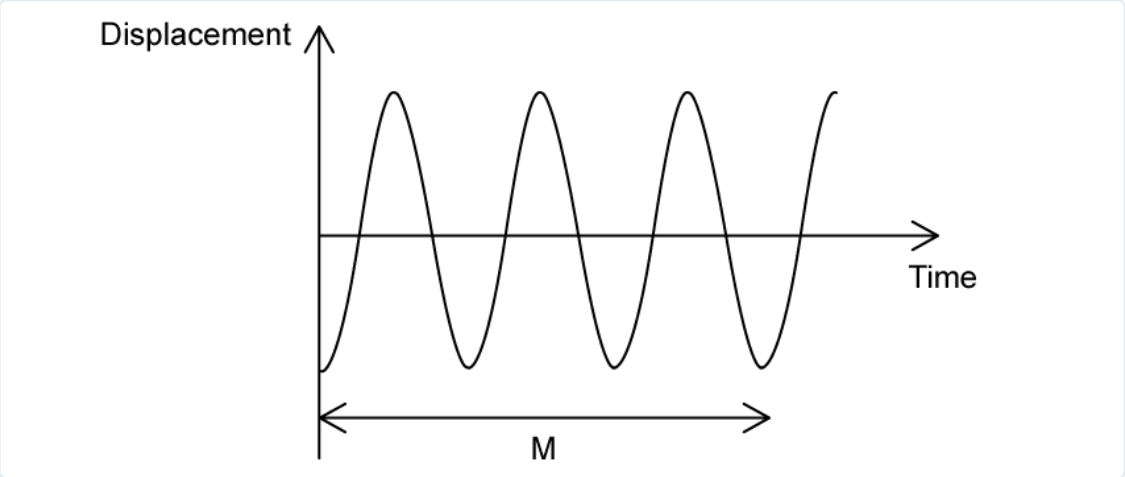


Fig. 1.4

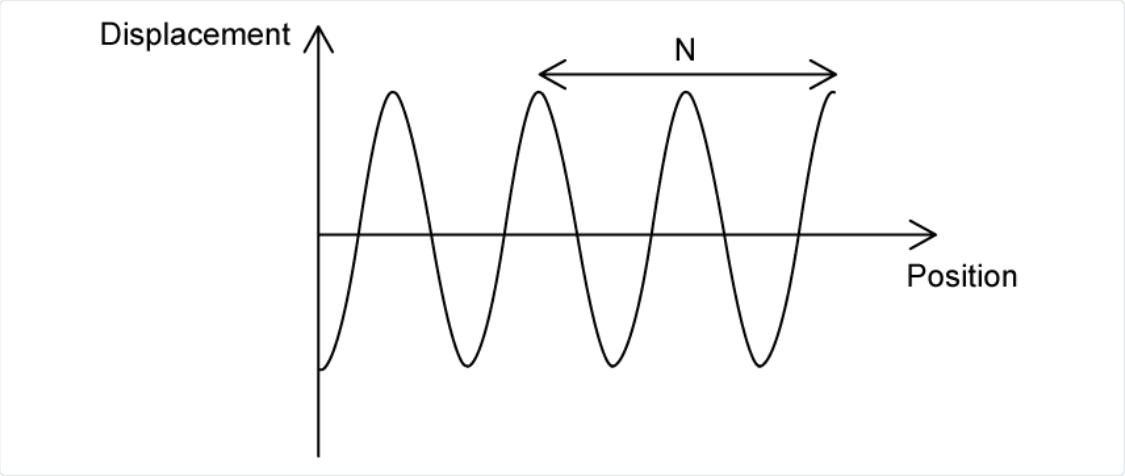


Fig. 1.5

Show that the wave speed *v* is given by:

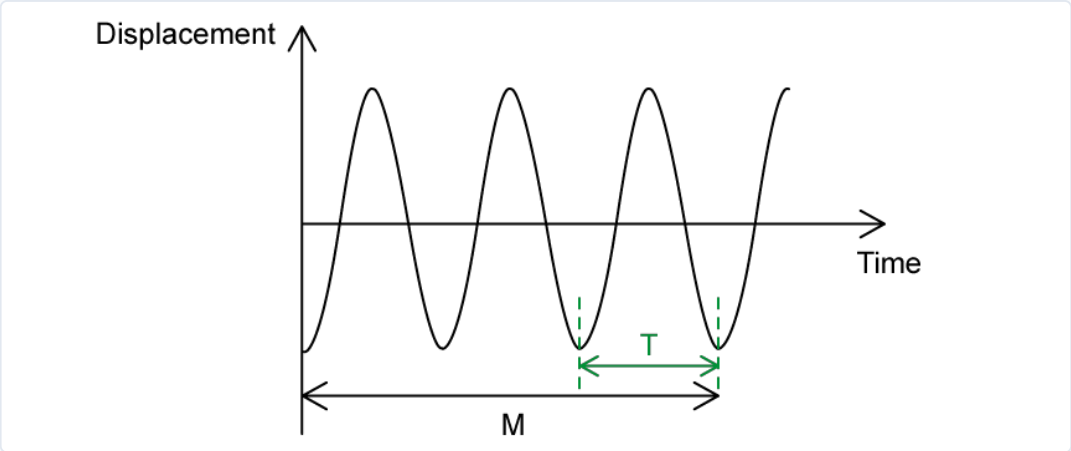
$$v = \frac{3N}{2M}$$

Mark Scheme (4 marks):

Show that the wave speed *v* is given by $v = \frac{3N}{2M}$:

Use Fig. 1.4 to write an expression for the time period, *T*:

- The time period is the time taken for one complete wave cycle
- This is determined using the time axis in the time-view of the wave, shown below:



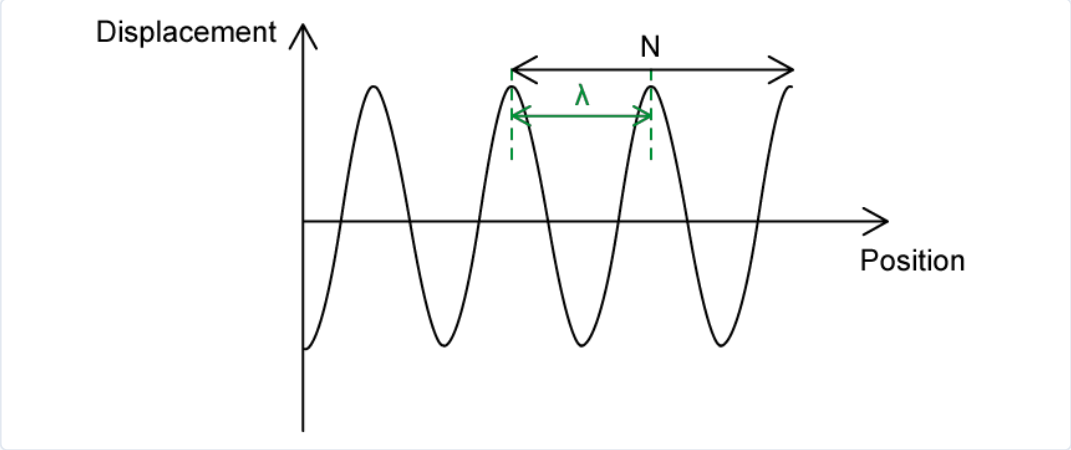
- Therefore, $T = \frac{M}{3}$ *[1 mark]*

Write an expression for the frequency *f* of the wave:

- $f = \frac{1}{T}$
- Therefore $f = \frac{1}{\frac{M}{3}} = \frac{3}{M}$ *[1 mark]*

Use Fig. 1.5 to write an expression for the wavelength λ of the wave:

- The wavelength of a wave is the distance between two corresponding points, e.g., peak to peak
- This is determined using the position axis in the space-view of the wave, shown below:



- Therefore, $\lambda = \frac{N}{2}$ *[1 mark]*

Write an equation for the wave speed *v* using the wave speed equation:

- $v = f\lambda = \frac{3}{M} \times \frac{N}{2} = \frac{3N}{2M}$ *[1 mark]*

[Total: 4 marks]

Question 6 Hard [13 marks]

Part (a) [2 marks]

Ultrasound is used to measure the depth of oceans, seas and lakes.

Fig. 1.1 shows a pulse of ultrasound being emitted from the boat, travelling down to the sea bed and being reflected back to the boat.

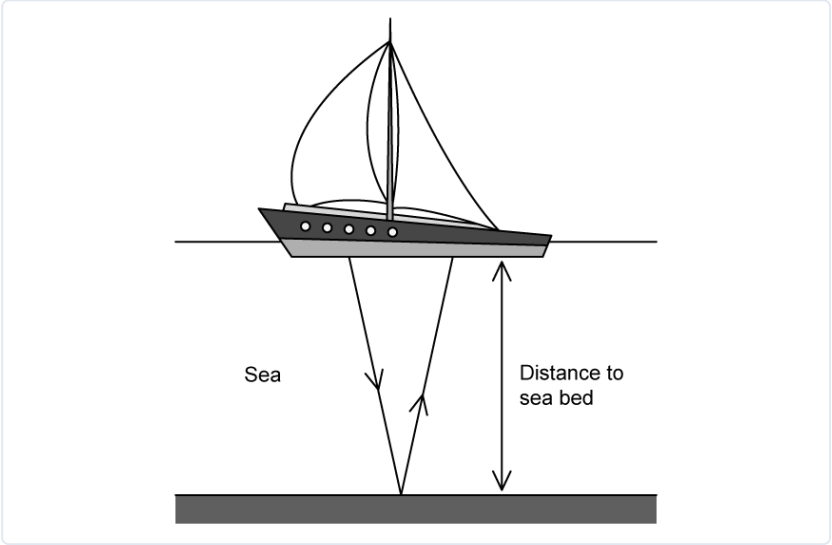


Fig.1.1

State which type of wave ultrasound is and explain your answer.

Mark Scheme (2 marks):

State which type of wave ultrasound is and explain your answer:

- Ultrasound is a sound wave (with a frequency above the range of human hearing); *[1 mark]*
- Therefore, ultrasound is a longitudinal wave; *[1 mark]*

[Total: 2 marks]

Part (b) [3 marks]

A cathode-ray oscilloscope (C.R.O.) is used to trace the ultrasound pulses sent from the boat and the reflected pulses returning to the boat. The C.R.O. trace is shown in Fig. 1.2.

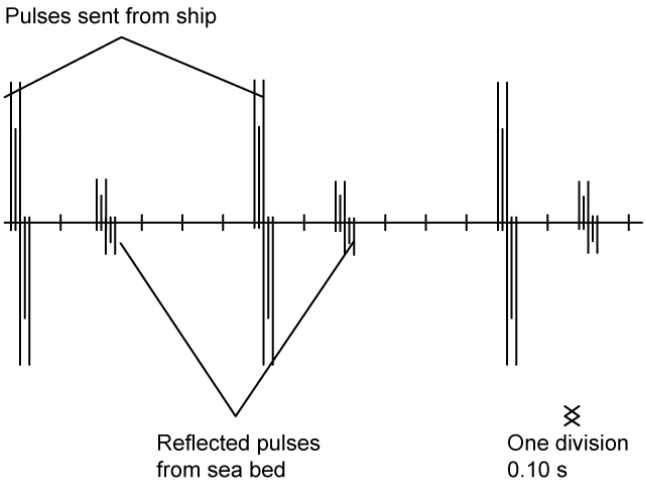


Fig. 1.2

The ultrasound travels through water at 1452 m s^{-1}, and the wavelength of the pulse is 0.023 m.

For the ultrasound pulses:

(i) Calculate the frequency

[1]

(ii) Calculate the distance to the sea bed

[2]

Mark Scheme (3 marks):

(i) For the ultrasound pulses, calculate the frequency, *f:*

List the known quantities:

- Speed of ultrasound wave in water, v = 1452 m s^{-1}
- Wavelength of ultrasound pulse, λ = 0.023 m

Using the wave speed equation:

- $v = f\lambda \rightarrow f = \frac{v}{\lambda}$
- $f = \frac{1452}{0.023}$ = 63130 Hz *[1 mark]*

(ii) For the ultrasound pulses, calculate the distance to the sea bed:

From the graph determine the time taken for the pulse to reach the sea bed:

- The tall pulses are those sent from the boat, the small pulses are those reflected from the sea floor
- The time between the tall emitted pulse and the smaller echo is two divisions of the axis
    - Each division of the axis is 0.10 s

- Therefore, the time between sending and receiving the pulses is 0.20 s

- However, the pulse travels from the boat to the sea bed and back in this time
  - Therefore, the time taken to reach the sea bed is 0.10 s
  - t = 0.10 s *[1 mark]*
- $v = \frac{d}{t}$
  - Therefore, $d = vt$
  - d = 1452 × 0.10 = 145.2 m *[1 mark]*

[Total: 3 marks]

Part (c) **[8 marks]**

The boat moves out to an area where the sea is deeper.

(i) State and explain two changes that would occur on the cathode-ray oscilloscope trace. You may include diagrams in your answer.

[4]

(ii) When the sea is over 450 m deep, the pulses must be transmitted less frequently. Explain why this is the case.

[4]

Mark Scheme (8 marks):

(i) State and explain two changes that would occur on the cathode-ray oscilloscope trace:

Change 1

State change 1:

- The horizontal separation between the transmitted pulse and the reflected pulse will increase; *[1 mark]*

Explain change 1:

- The sea is deeper so the distance travelled by the pulses increases, and will therefore take longer to reach the sea bed, (and longer to return); *[1 mark]*

Change 2

State change 2:

- The amplitude of the reflected pulses will be smaller / decrease; *[1 mark]*

Explain change 2:

- Some of the energy is absorbed by the surrounding water. Since they have to travel through a greater depth of water, more energy will be lost; *[1 mark]*

(ii) When the sea is over 450 m deep, the pulses must be transmitted less frequently. Explain why this is the case

Calculate the time taken between the transmission and detection of each pulse at 450 m:

- $t = \frac{d}{v} = \frac{450 \times 2}{1452}$ = 0.6 s *[1 mark]*

Explain why this means the pulses must be transmitted less frequently:

- The cathode-ray oscilloscope trace shows that the pulses are transmitted every 0.6 s; *[1 mark]*
- At a depth of 450 m, the reflected pulses would coincide with / arrive at the same time as the transmitted pulses; *[1 mark]*
- At greater depths, the reflected pulses would be detected after subsequent / the next pulses have been transmitted; *[1 mark]*

[Total: 8 marks]